

CARMEN QUINONES

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EDUCATION

California Institute of Technology (Caltech)

Pasadena, CA

B.S. Mechanical Engineering, Minor in Aerospace Engineering and Robotics **GPA: 3.8**

Expected June 2028

Relevant Coursework: Thermodynamics, Solid and Fluid Mechanics, Robotics Laboratory, Mechanical Prototyping & Design, CNC Machining, Nanofabrication Laboratory

SKILLS

- **Design & Fabrication:** HASS CNC Machining, Nanofabrication (SEM, Photolithography), CAD/CAM (SolidWorks, OnShape), Waterjet, Laser Cutting, Rapid Prototyping, GD&T, BOMS
- **Automation & Controls:** SCADA (Ignition, Tulip), PLCs & HMIs (Studio 5000, TIA Portal), Factory Acceptance Testing (FAT), Breadboarding, Arduino
- **Programming:** Python, Java, MATLAB, SQL

WORK EXPERIENCE

Dynamic Motion Control, Chicago, IL: *Automation Engineering Intern*

Summers 2025 – 2026

- Executed Factory Acceptance Testing (FAT) for 20+ industrial control units while building custom API endpoints, UI views, and interactive screens in Ignition
- Automated work instructions and data logging for active production lines, streamlining build traceability across 1,000+ manufactured units.
- Programmed PLCs and HMIs across Rockwell Studio 5000 and Siemens TIA Portal to commission reliable automation systems onsite.

Northwestern McCormick School of Engineering, Evanston, IL: *Researcher*

July 2026 – September 2026

- Analyzed literature on ergodic robotic control and controlled diffusion, critically to provide actionable editorial feedback.
- Designed and fabricated a specialized robotic gripper integrated with custom sensors to enable advanced robotic machine learning models

Caltech ME13/113, Pasadena, CA: *Undergraduate Teaching Assistant*

January 2026 – March 2026

- Individually instructed over 30 students in precision metalworking
- Guided the full machining process from drawing interpretation and tool selection to fabrication and prototyping.

Caltech Air Outer Space Team, Pasadena, CA: *Researcher*

June 2024 – January 2025

- Prototyped and developed autonomous landing structure published report in AIAA on testing readiness levels and methodologies. Project was recognized as a finalist in the NASA 2024 BIG Idea Challenge.

PROJECTS

RC Rocker-Bogie Rover, Caltech

June 2026

- Engineered a rocker-bogie suspension system to traverse rough terrain with minimal body vibration, integrated a 4-DOF robotic arm with a custom gripper and, R/C-controlled through Arduino and 2x45A RoboClaw motor controllers

Bike Wheel Transmission, Caltech

May 2026

- Co-designed and fabricated a custom belt-drive transmission in a laser-cut acrylic gearbox under a \$200 budget.
- Placed 4th overall out of all competing teams by optimizing timing belts, pulleys, and shafts for maximum power transfer and efficiency.

ACTIVITIES

Associated Students of Caltech (ASCIT), Pasadena, CA: *Director of Operations*

June 2025 – Present

- Serve as liaison to publication officers for Caltech's primary student government body.
- Direct oversight of all Corporation property, maintaining operational files and liaising with all student organizations.

Caltech Hispanic and Latino Association, Pasadena, CA: *Vice President & Treasurer*

June 2025 – Present

- Oversee club finances, budgeting, and funding initiatives; secured over \$15,000 in grants and sponsorships to support cultural and community programs.

HONORS & AWARDS:

- QuestBridge National College Match Scholarship, The Gates Scholarship Recipient